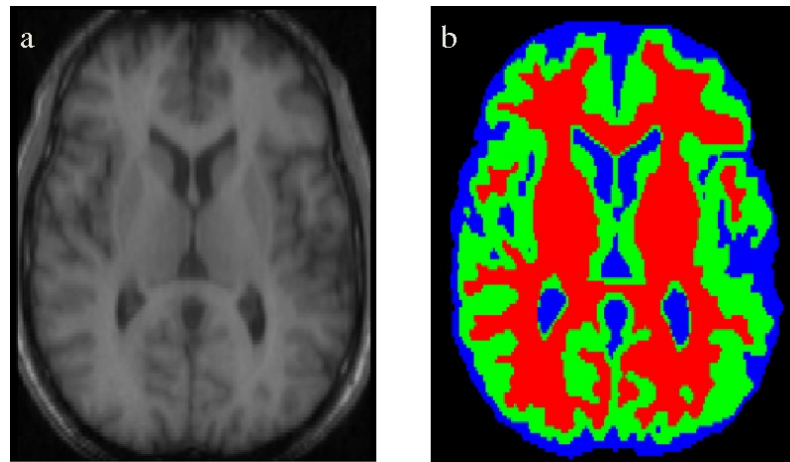


Generative AI and Foundation Models for Medical Image Segmentation in Ultra-Low Data Regimes

Pengtao Xie

UC San Diego

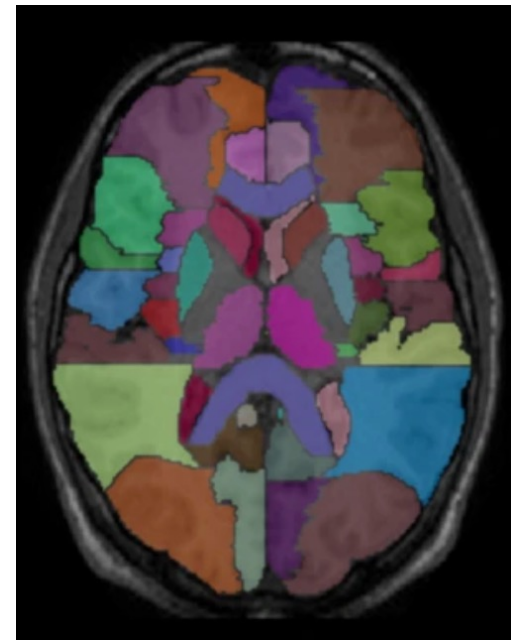
Medical image semantic segmentation



- Broad applications in disease diagnosis and progression tracking, treatment planning, surgery assistant, etc.

Deep learning for semantic segmentation

- Strong performance
 - UNet, DeepLab, SegFormer, etc.
- Need large amounts of labeled training data
- Segmentation masks of medical images are difficult to label
 - Require strong domain expertise
 - Objects in medical images have complex structure
 - Medical images have large size



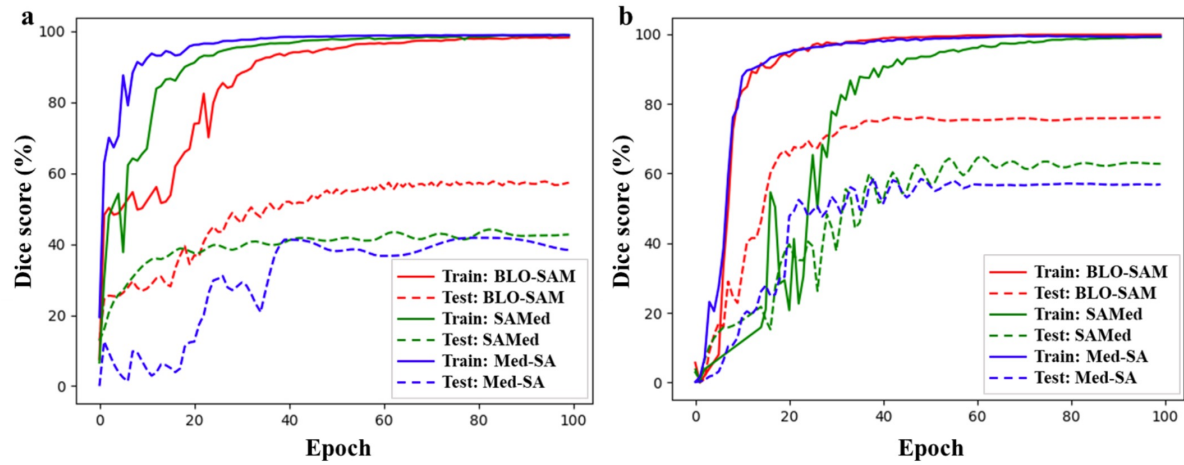
Overfitting-Preventing Finetuning of the Segment Anything Model

BLO-SAM: Bi-level Optimization Based Finetuning of the Segment Anything Model for Overfitting-Preventing Semantic Segmentation

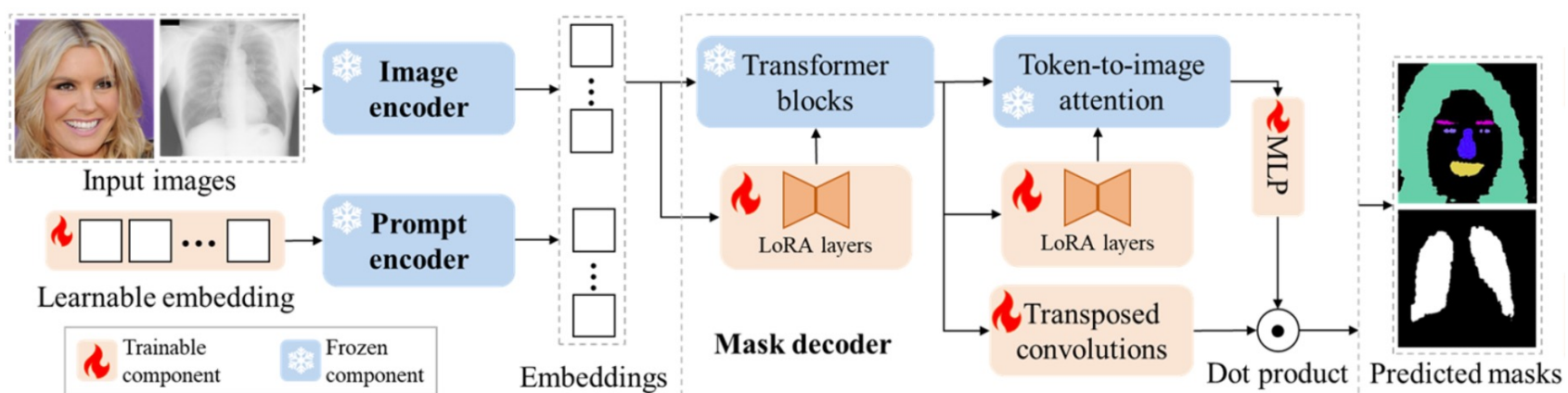
Li Zhang, Youwei Liang, Ruiyi Zhang, Amirhosein Javadi, Pengtao Xie

ICML 2024

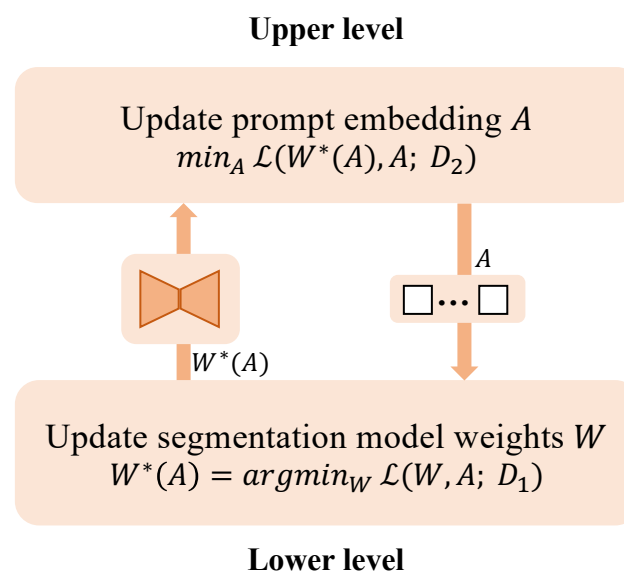
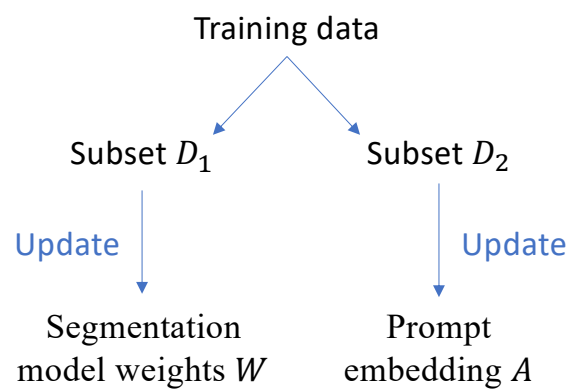
Overfitting of SAM baselines



Overview



Overview



Bi-level optimization

- Lower level

$$W^*(A) = \arg \min_W \mathcal{L}(W, A; D_1)$$

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{CE}(W, A; D_1) + \lambda\mathcal{L}_{Dice}(W, A; D_1)$$

Bi-level optimization

- Upper level

$$\min_A \mathcal{L}(W^*(A), A; D_2).$$

Bi-level optimization

$$\begin{array}{ll} \min_A & \mathcal{L}(W^*(A), A; D_2) \\ \text{s.t.} & W^*(A) = \arg \min_W \mathcal{L}(W, A; D_1) \end{array}$$

Results

Method	Teeth		Kvasir		Lung		Total	Trainable	Train Cost	Manual Prompts
	4 examples	8 examples	4 examples	8 examples	2 examples	4 examples	Param(M)	Param(M)	(GPU hours)	
DeepLab	56.8	63.9	31.9	37.8	61.1	76.4	41.9	41.9	0.27	✗
SwinUnet	28.6	50.6	37.8	39.0	62.1	76.4	27.2	27.2	0.29	✗
HSNet	70.2	72.2	30.0	35.1	83.1	84.5	28.1	2.6	0.16	✗
SSP	33.7	51.7	27.4	27.7	83.2	90.1	8.7	8.3	0.22	✗
SAM	21.2	21.2	14.7	14.7	31.4	31.4	91.0	0	0	✓
Med-SA	69.8	75.1	33.5	59.3	82.9	91.7	104.4	10.7	0.62	✓
SAMed	69.9	76.4	42.7	57.1	84.4	91.8	91.0	1.1	0.46	✗
BLO-SAM	73.2	77.3	59.7	61.6	87.1	93.7	91.0	1.1	0.57	✗

Dice scores

Results

Method	Training with 4 labeled examples						Training with 8 labeled examples						Manual Prompts
	Overall	Brow	Eye	Hair	Nose	Mouth	Overall	Brow	Eye	Hair	Nose	Mouth	
DeepLab	49.5	33.2	55.9	55.2	55.8	47.5	54.9	37.3	59.7	58.0	64.0	55.7	✗
SwinUnet	35.2	21.7	21.7	46.7	44.4	41.6	42.4	28.8	33.1	52.7	47.6	49.8	✗
HSNet	49.9	30.1	41.9	58.9	59.6	59.0	60.1	43.6	58.1	76.6	58.2	64.2	✗
SSP	56.9	40.3	33.6	73.2	71.6	65.6	60.0	45.4	31.2	76.6	74.0	72.7	✗
SAM	32.9	20.8	30.6	44.0	44.1	25.2	32.9	20.8	30.6	44.0	44.1	25.2	✓
Med-SA	62.9	36.3	63.6	77.6	71.0	66.0	67.7	42.9	65.5	82.4	74.6	73.1	✓
SAMed	58.2	28.3	55.9	78.5	65.1	63.0	65.0	39.5	66.0	82.4	70.5	66.4	✗
BLO-SAM	65.9	39.4	65.5	82.8	74.3	67.6	69.9	45.8	71.1	83.6	76.1	72.7	✗

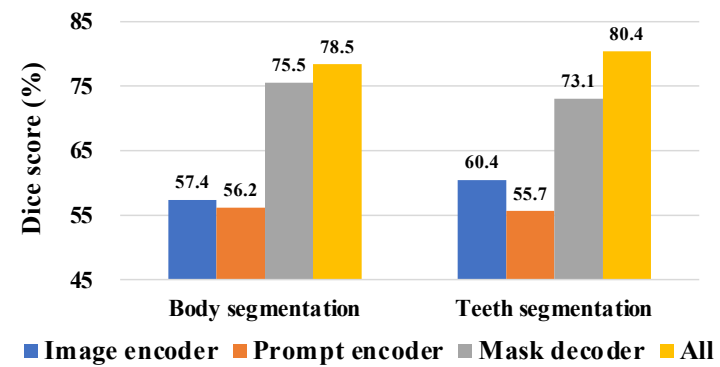
Dice scores

Results

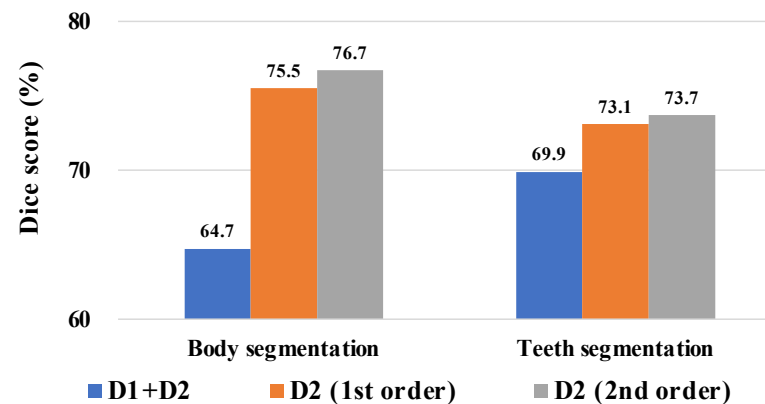
Method	Car components								Human body	
	Training with 2 labeled examples				Training with 4 labeled examples				4 examples	8 examples
	Overall	Body	Wheel	Window	Overall	Body	Wheel	Window		
DeepLab	46.5	58.5	55.0	26.1	59.8	62.9	66.6	49.8	31.8	37.0
SwinUnet	31.4	22.2	33.4	38.7	47.3	49.3	53.7	38.9	30.9	56.8
HSNet	51.0	67.1	32.4	53.4	68.8	70.8	70.3	65.4	50.6	55.8
SSP	64.1	62.8	76.2	53.3	72.2	77.7	78.2	60.6	58.9	76.5
SAM	35.1	40.8	41.7	22.9	35.1	40.8	41.7	22.9	26.0	26.0
Med-SA	67.3	80.8	70.9	50.3	75.9	84.4	78.1	65.3	58.8	80.6
SAMed	60.4	78.8	65.0	37.5	74.0	85.3	72.5	64.2	63.8	81.5
BLO-SAM	71.1	83.2	74.5	55.6	78.3	86.3	79.9	68.6	76.3	85.1

Dice scores

Ablation study of different trainable components



Ablation study on approaches for optimizing the prompt embedding



Summary

Approach

- Learn the prompt embedding and model parameters of SAM on two distinct sub-datasets.
- Bi-level optimization.

Advantages

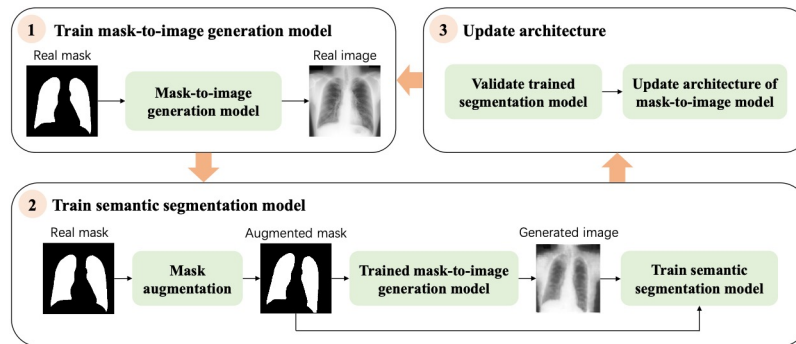
- Resilient to overfitting.
- Fully automated without requiring manual prompts.

Results

- Strong segmentation performance with less than ten training examples.
- Significantly outperform vanilla SAM and its variants, few-shot segmentation methods, and popular segmentation models.

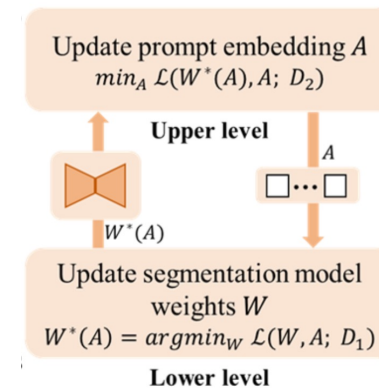
Conclusions

End-to-end data generation



Medical image segmentation in ultra-low data regimes

Overfitting-preventing finetuning of SAM



Thank you!

Questions?